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Original Empirical Article

An 18-Year Study of Changes in Neurocognitive Function and Associations with Repetitive Head Trauma among Former Collegiate American Football Players: A Case Series

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ABSTRACT

Objectives: To assess changes in neurocognitive test scores among former collegiate American football players over 18 years and examine associations with head trauma history.

Methods: Former collegiate American football players ($n = 31$; aged $= 38.4 \pm 1.3$ years) self-reported their concussion history and repetitive head impact exposure (Head Impact Exposure Estimate). Neurocognitive testing was conducted at two time-points (during college [T1] and 18-year follow-up [T2]) via Hopkins Verbal Learning Test–Revised Immediate and Delayed Recall; Verbal fluency; Symbol Digit Modalities Test; and Trail-Making Test-B. Raw score changes were calculated, with accompanying Cohen's *d* effect sizes and coefficients of variation. Repeated-measures analyses of covariance models were fit to examine changes in test performance. Multivariable linear regression models tested associations of sport-related concussion history and repetitive head impact exposure with change scores.

Results: No significant changes in cognitive test performance were observed ($ps \geq .06$). Individual-level changes exhibited high variability (coefficients of variation $\geq 244\%$), and group-level effects were small to medium ($d \leq 0.41$). Neither sport-related concussion history nor repetitive head impacts were related to change scores ($ps > .05$).

Conclusions: Group-level test scores did not change over 18 years among former collegiate football players now in midlife, though individual-level variability was high. Sport-related concussion and head impact exposure estimates were not related to change. Longitudinal studies are essential to understand cognitive trajectories of former football players and factors influencing those trajectories.

Keywords: Brain health; Cognition; Concussion; Repetitive head impacts; Athlete health

INTRODUCTION

Among former American football players, history of one or more sport-related concussions has been associated with subjective (i.e., self-reported) cognitive difficulties but not necessarily with objectively measured cognitive test performance (Brett et al., 2021b; Bryant et al., 2022; Guskiewicz et al., 2005; Walton et al., 2021a; Wright et al., 2016). Conversely, history of multiple concussions has also been associated with higher prevalence of cognitive disorders and dementia (Guskiewicz et al., 2005; Walton et al., 2022a). There is mixed evidence regarding the relationship of lifetime exposure to repetitive head impacts (RHIE), defined by repeated blows to the head that do not result in overt clinical signs and symptoms of brain injury, with both self-reported cognitive complaints and objective clinical measures of cognition (e.g., neurocognitive tests) (Alosco et al., 2019; Brett et al., 2022a; Bryant et al., 2022; Fields et al., 2020; Montenigro et al., 2017; Schaffert et al., 2022, 2024). Most research with former football players has been cross-sectional in design, and, despite the lack of longitudinal investigations into the association between sport-related head trauma and later-life cognitive outcomes, current and former American football players have expressed concerns about their long-term cognitive health (Alosco et al., 2023; Baugh et al., 2021; Walton, Kerr, Mannix, et al., 2021b). Therefore, it is important to measure changes in cognitive function as former football players age, as well as factors relating to measurable changes, in order to improve understanding of long-term cognitive health in this population.

There is a large amount of research examining the long-term brain health of former professional American football players; however, the body of literature examining the brain health of

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former players in relation to lower levels of competitive play (e.g., collegiate) is smaller and retrospective (Phelps et al., 2022). There is also limited evidence for brain health outcomes among middle-aged former players—a time period that is especially important because it can provide insight into the course and potential development of cognitive problems prior to the typical period of onset for cognitive and dementia-related disorders. Existing studies of middle-aged former American football players have been cross-sectional in design, which limits our ability to draw inferences about the trajectories of cognitive functioning and which factors may affect these trajectories as former players age. Longitudinal studies, particularly those with former players at mid-life, are essential to understand better how concussions and RHIE may influence trajectories of cognition many years after sport discontinuation.

The primary objectives of this preliminary longitudinal study were to: (A) examine change in neurocognitive test performances from baseline assessment in college to follow-up testing in early midlife among former collegiate American football players and (B) examine whether history of head trauma (concussion history and RHIE) was associated with changes in neurocognitive test performance. An exploratory aim of this study was to test for relationships between changes in neurocognitive test performance and a number of additional football-related (e.g., played professionally) and health-related (e.g., depressive symptoms) factors. Results from this preliminary study will be helpful to inform the development of future clinical research studies of collision-sport athletes, particularly with prospective, longitudinal methods to address concerns related to head trauma and long-term brain health.

MATERIALS AND METHODS

Participants

Participants were former collegiate football players ($n = 31$) who were approximately 15 years removed from collegiate sport participation. These individuals had participated in the original National Collegiate Athletics Association (NCAA) Concussion Study (August 1, 1999–January 3, 2002) (Guskiewicz et al., 2003; McCrea et al., 2003), as well as a 15-year follow-up study (July 26, 2016–August 3, 2018) (Brett et al., 2022a; Kerr et al., 2018). The original NCAA Concussion Study and the 15-year follow-up study were each approved by the Institutional Review Boards at both the University of North Carolina at Chapel Hill and the Medical College of Wisconsin. All participants provided written informed consent prior to participation in each of the studies. To be eligible for the original NCAA Concussion Study, participants needed to have completed pre-injury baseline assessments as part of their routine clinical care at one of the participating NCAA member institutions (Guskiewicz et al., 2003; McCrea et al., 2003). To be eligible for the follow-up study, participants must have participated in at least 1 year of collegiate football. Potential participants who reported a history of psychotic disorder with active symptoms, any contraindications to participation in the study (e.g., preclusion to MRI), or were unable to travel to one of the two research sites were excluded from the follow-up study. For the present analyses, 15-year follow-up study participants who were missing neurocognitive testing data due to not participating in the original NCAA concussion study were excluded ($n = 34$).

Outcome data for our analyses originated from in-person neurocognitive testing assessments at both the baseline (T1; while playing collegiate football; c.1999–2001) and longitudinal follow-up (T2; approximately 15 years after collegiate football play; c.2016–2018). In addition, survey data at T2 recorded concussion and RHIE histories and additional exploratory factors.

Neurocognitive tests

Participants completed a battery of paper-and-pencil neurocognitive tests at each time point that were administered by a member of the research team. Research team members were trained in the administration of these tests by a licensed neuropsychologist (e.g., MAM). The neurocognitive test battery included the Hopkins Verbal Learning Test–Revised (HVLT-R) (Brandt & Benedict, 2001), FAS Verbal Fluency (FAS) (Spreen & Benton, 1977), the Symbol Digit Modalities Tests (SDMT) (Smith, 2007), and the Trail Making Test–Form B (TMT-B) (Reitan & Wolfson, 1985). A measure of single word reading (Wechsler Test of Adult Reading; WTAR) was also completed and used as an estimate of premorbid abilities (Wechsler, 2001).

Concussion history & Head Impact Exposure Estimate

On a questionnaire at T2, each participant was provided an operational definition of concussion used in previous research with this population (Kerr et al., 2018; McCrea et al., 2004): "an injury occurring typically, but not necessarily, from a blow to the head, followed by a variety of symptoms that may include any of the following: headache, dizziness, loss of balance, blurred vision, 'seeing stars,' feeling in a fog or slowed down, memory problems, poor concentration, nausea, throwing up, and loss of consciousness." Participants then subsequently self-reported their lifetime concussion history (SR-CHx), including injuries sustained while participating in sport or through other mechanisms (e.g., motor vehicle accidents, military service). This method of self-reported concussion history has shown moderate consistency across multiple repetitions separated by multiple years in former football players (Kerr et al., 2022a). Self-reported lifetime concussion history was then separated into four categories: 0–1; 2–4; 5–7; and 8+ in congruence with prior studies with this same sample of participants (Brett et al., 2021a; Bryant et al., 2022; Walton et al., 2022b).

Estimation of RHIE through organized football participation was estimated using a survey and semistructured interview data that have been described previously (Brett et al., 2022a; Kerr et al., 2015). Briefly, the Head Impact Exposure Estimate (HIEE) was derived from the players' self-reported participation in football games and practices across each individual year of play in the high school, collegiate, and professional settings (Kerr et al., 2015). Hours of participation in contact practices and games were adjusted for playing position(s), phase of the season (e.g., preseason, regular season, postseason), and the proportion of each game played for each position at each individual level of play according to estimated numbers of head contacts derived from helmet accelerometer studies (Broglio et al., 2011; Crisco et al., 2010; Kerr et al., 2015). The resultant HIEE metric was used

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as a surrogate for RHIE during football participation from high school and afterward.

Exploratory factors

Exploratory variables were selected *a priori* based on their potential capacity to affect cognition and acquired via the self-administered questionnaire at T2. Self-reported depression, anxiety, and somatic symptoms were reported on the Brief Symptom Inventory-18 (BSI-18) (Derogatis, 2001). Sleep quality was measured with the Pittsburgh Sleep Quality Index Total Score (PSQI) (Buysse et al., 1989). Measures of alcohol use (Alcohol Use Disorders Identification Test; AUDIT) (Bush, 1998), and drug abuse (Drug Abuse Screening Test; DAST) (Gavin et al., 1989) were also included. Total years of participation in competitive football (TYP) was also reported by participants as it has been previously used as a surrogate for RHIE separate from HIEE (Brett et al., 2022a).

Analyses

Raw neurocognitive test outcome scores were used in all inferential analyses. Raw change scores were calculated by subtracting outcome scores for each test at T1 from outcome scores at T2, and Cohen's *d* effect sizes were calculated alongside coefficients of variation (CoVs) to assess the magnitude and variability of test performance differences between timepoints, respectively. Cohen's *d* effect sizes were interpreted as small (0.20–0.49), medium (0.50–0.79), or large ($\geq$0.8) (Cohen, 1988). Repeated-measures analysis of covariance (ANCOVA) models were fit to test for significant differences in test performance over time while accounting for estimated premorbid abilities as reflected in standardized WTAR scores. Additionally, we calculated the proportion of the sample whose performance was more than or equal to 1.5 *SD* below the normative mean for each measure at both time points, accounting for age and current level of education at the time of test administration, based on available normative data (Brandt & Benedict, 2001; Tombaugh, 1999, 2004).

Subsequently, multivariable linear regression models were fit to test the associations of SR-CHx and HIEE with neurocognitive test change scores while controlling for premorbid abilities (i.e., WTAR). Standardized beta-values (β) and *p*-values were computed for SR-CHx and HIEE in each regression model. As a separate sensitivity analysis, SR-CHx was operationalized as a continuous variable, rather than categorical, and included in the models alongside HIEE and WTAR scores.

Due to the non-normal distribution of variables included in our exploratory measures, bivariate Spearman's Rho (ρ) correlation analyses were performed to examine associations between exploratory variables (BSI-18 depression, anxiety, and somatic symptoms; PSQI total score; AUDIT score; DAST score; and TYP) and change scores for each cognitive test. All analyses were performed with SPSS version 28.0 (Armonk, NY, USA), and *a priori* α was set to 0.05.

RESULTS

Data from all 31 eligible former collegiate football players were included in this study (Table 1). The majority of former players identified as White ($n = 25$; 80.6%), three (9.7%) identified as Black or African American, and three (9.7%) identified as two or more races. A substantial minority ($n = 14$; 45.2%) obtained a graduate or professional degree, and five (16.1%) reported playing football at a professional level (Table 1). At T1, the number of participants whose performance fell at or more than 1.5 *SD* below the normative mean across measures were as follows: HVLT-R Immediate Recall ($n = 13$ [41.9%]), HVLT-R Delayed Recall ($n = 12$ [38.7%]), FAS ($n = 1$ [3.2%]), SDMT ($n = 0$ [0%]), and TMTB ($n = 2$ [6.5%]). At T2, a smaller proportion of participants' scores fell at or 1.5 *SD* below the normative mean for HVLT-R Immediate Recall ($n = 3$ [9.7%]), HVLT-R Delayed Recall ($n = 2$ [6.5%]), FAS ($n = 1$ [3.2%]), and SDMT ($n = 0$ [0%]), with the exception of TMTB ($n = 7$ [22.6%]).

Changes in neurocognitive test performance over time

Participants completed their second study visit (T2) an average of 18.2 ± 0.8 years after their initial study participation while playing collegiate football (T1). Neurocognitive test raw score differences were highly variable from T1 to T2, with CoV values ranging between 244% and 560%, and small-to-medium effect sizes for the differences between time points (Table 2). Individual participant raw score differences for each neurocognitive test ranged from improvements to declines in test performance over time, as depicted in Fig. 1. While controlling for estimated premorbid abilities, there were no significant changes over time in neurocognitive test performance at the group level (*p*-values $\geq$ .06; Table 2). For FAS, the one measure in which a trend association was observed ($p = .06$), the direction of change from T1 to T2 reflected improvement in performance.

History of head trauma and changes in neurocognitive test performance

Individual change scores for each participant are presented in relation to HIEE and SR-CHx in Fig. 1. While controlling for estimated premorbid abilities, there were no statistically significant associations between SR-CHx and change on any of the neurocognitive test outcomes over time (*p*-values > .05)—regardless of the operationalization used for concussion history (categorical or continuous; Table 3). Similarly, HIEE was not significantly associated with change in any of the neurocognitive test outcomes over time (*p*-values >.16).

Exploratory predictors of change in neurocognitive test performance

Significant correlations were observed for the relationship between BSI-18 Somatic symptom severity at T2 and both HVLT-R Immediate Recall ($\rho = -0.395$; $p = .028$) and SDMT ($\rho = -0.380$; $p = .035$) change scores (Table 4). These associations indicated that greater somatic symptom burden at T2 was related to larger declines in test performance over time. Additionally, higher DAST scores at T2 were also significantly correlated with larger declines in HVLT-R Immediate Recall ($\rho = -0.370$; $p = .040$; Table 4). There were no significant correlations observed between cognitive test change scores and depression symptoms, anxiety symptoms, sleep quality, scores on a screening measure of hazardous alcohol use, or total years of football play.

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Table 1. Participant information ($n = 31$)

Age in years at baseline (T1), mean (SD)[a]	20.2 (1.3)
Age in years at follow-up (T2), mean (SD)	38.4 (1.3)
Racial identity, *n* (%)	
White	25 (80.6)
Black or African American	3 (9.7)
Two or more racial identities selected	3 (9.7)
Highest degree completed, *n* (%)	
Undergraduate degree (e.g., BA, BS)	17 (54.8)
Graduate or professional degree (e.g., MEd, MS, PhD, JD)	14 (45.2)
Years between baseline (T1) and follow-up (T2), mean (SD)	18.2 (0.8)
Total years playing football, mean (SD)	11.9 (3.5)
Played professional football, *n* (%)	
Yes	5 (16.1)
No	25 (80.6)
Not reported	1 (3.2)
Years since sport participation, mean (*SD*)	15.8 (1.7)
Self-reported concussion history, *n* (%)	
0–1	10 (32.3)
2–4	6 (19.4)
5–7	7 (22.6)
8+	8 (25.8)
Head Impact Exposure Estimate, mean (*SD*)	1318.0 (550.4)
Wechsler Test of Adult Reading (WTAR)[b], mean (*SD*)	111.3 (8.3)
Brief Symptom Inventory-18 (BSI-18) domain score, raw mean (*SD*)	
Depression	3.0 (4.3)
Anxiety	3.3 (3.8)
Somatic	2.8 (4.2)
Pittsburgh Sleep Quality Index (PSQI) Total Score, mean (*SD*)	6.4 (3.5)
Alcohol Use Disorders Identification Test (AUDIT) total score, mean (*SD*)	6.2 (5.2)
Drug Abuse Screening Test (DAST) total score, mean (*SD*)	0.2 (0.5)

[a]Current academic year in college for participants at baseline (T1) were: 1st year ($n = 3$), 2nd year ($n = 6$), 3rd year ($n = 8$), 4th year ($n = 11$), 5th year ($n = 2$), and academic year was missing for one participant. [b]WTAR standardized score was reported here and was included in analyses as a covariate.

Table 2. Neurocognitive test performance changes over time. Change in raw score was the simple difference between each neurocognitive test raw score at T2 compared to T1. Change score Cohen's *d* effect sizes were calculated using the change score standard deviations. Effects of time were modeled via ANCOVA models, with Wechsler Test of Adult Reading (WTAR) standard scores included as a covariate

	Change in raw score (T2 – T1) ***mean (sd)***	**Coefficient of variation** ***sd/mean***	**Change score effect size Cohen's *d* (95% CI)**	**Effect of time *p*-value**
HVLT-R Immediate Recall	1.68 (5.23)	312%	0.32 (0.00–0.68)	0.34
HVLT-R Delayed Recall	0.58 (2.62)	451%	0.22 (0.00–0.58)	0.38
FAS	4.35 (11.49)	264%	0.38 (0.01–0.74)	0.06
SDMT	−1.90 (10.66)	560%	0.18 (0.00–0.53)	0.80
TMT-B	−8.16 (19.89)	244%	0.41 (0.04–0.77)	0.44

Note. T1 = baseline, while playing collegiate football; T2 = follow-up time point; sd = standard deviation; HVLT-R = Hopkins Verbal Learning Test–Revised; FAS = F-A-S verbal fluency; SDMT = Symbol Digit Modalities Test; TMT-B = Trail-Making Test–Form B.

DISCUSSION

Evidence surrounding the long-term effects of concussions and RHIE in former athletes has primarily been reported in cross-sectional studies, some of which have also controlled for potential confounders (e.g., current symptoms, medical history). Our preliminary study represents an important step forward from cross-sectional investigations, whereby we explored a number of clinically salient factors in a cohort of American football players enrolled during their collegiate football participation with subsequent follow-up after nearly two decades. It is notable that while many former players exhibited declines in test performance between study visits, many also exhibited improvements (Fig. 1), with a high degree of inter-individual variability. These disparate functional trajectories warrant continued follow-up, including future studies with larger samples of former athletes that can account for the many potential determinants of brain health beyond head trauma exposure alone.

Change in cognitive test performance over time

We observed high variability for individual-level cognitive test performance differences across all test outcomes, with CoV scores indicating that the standard deviation for change scores ranged from 244% (FAS) to 560% (SDMT) of the mean value

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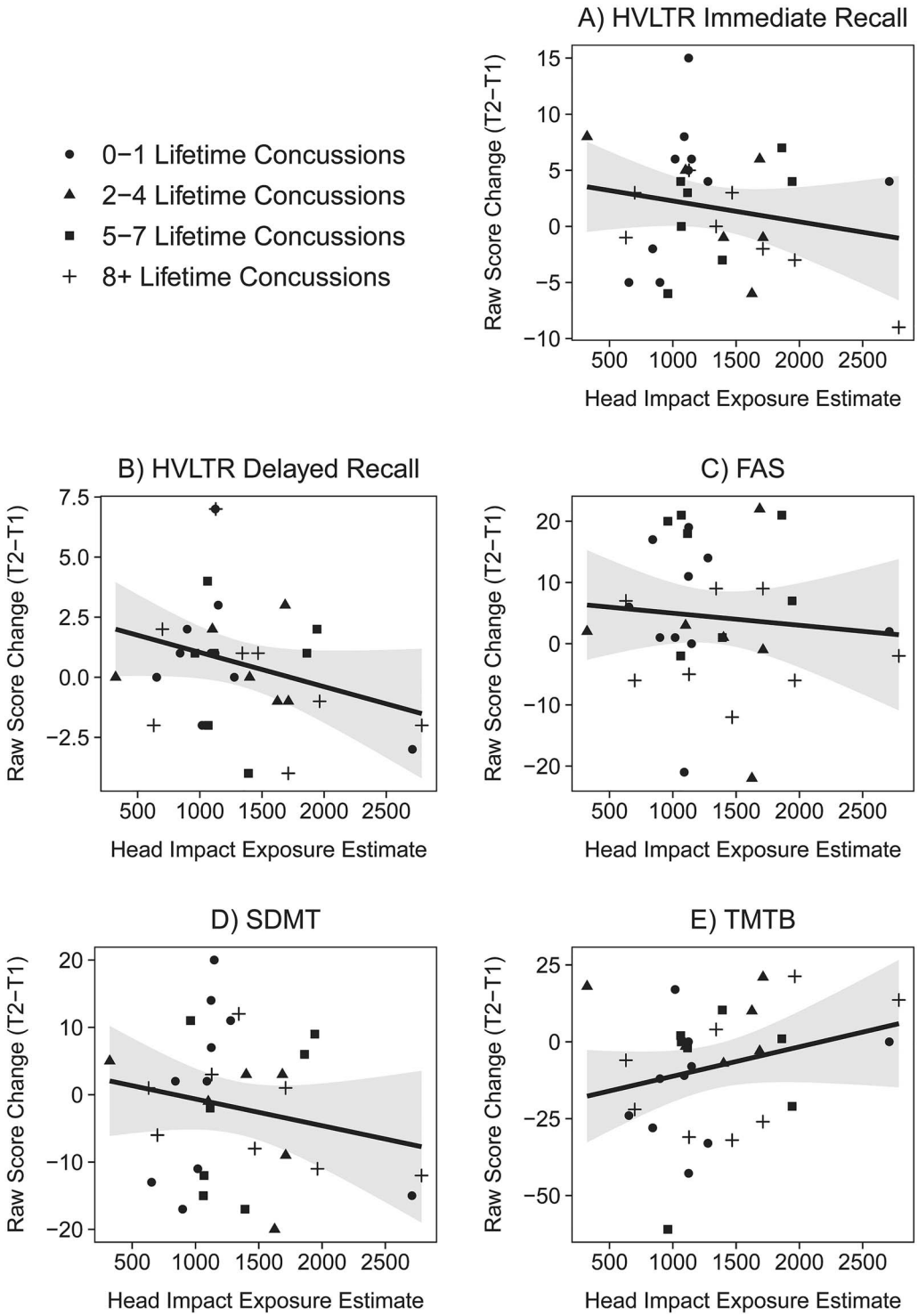

Fig 1. Head impact exposure estimates (HIEEs), self-reported concussion history (SR-CHx), and raw score changes in neurocognitive test performance.

for each change score (Table 2; Fig. 1). Both improvements and declines in performance were observed across 18 years for each test outcome score in this sample, suggesting that there was not a clear pattern of change among these former American football players in early midlife. This was further evidenced by small-to-medium effect sizes (Cohen's *d*) for the mean difference between test outcome scores at T1 and T2, and the lower bound of the 95% confidence intervals for all effect sizes approximated 0.0 (minimal to no effect observed; Table 2). Collectively, these findings suggest that person-specific factors are likely to influence individual functional trajectories, many of which we explored in this preliminary longitudinal study. If there are

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Table 3. Results from multivariable linear regression models testing the effects of head trauma history on change in neurocognitive test performance. Multivariable linear regression models were fit with self-reported concussion history (SR-CHx) and the Head Impact Exposure Estimate (HIEE) as the predictor variables, and WTAR was included as a covariate. As a sensitivity analysis, SR-CHx was operationalized into four categories for the initial set of models and as a continuous variable in a separate (sensitivity) set of models

	Categorical SR-CHx		Continuous SR-CHx	
	β	*p*-value	β	*p*-value
HVLT-R Immediate				
SR-CHx	−0.262	.17	−0.385	.05[a]
HIEE	−0.114	.57	−0.019	.93
HVLT-R Delayed				
SR-CHx	−0.049	.80	−0.042	.84
HIEE	−0.272	.19	−0.265	.22
FAS				
SR-CHx	−0.063	.75	−0.129	.52
HIEE	−0.082	.67	0.091	.66
SDMT				
SR-CHx	−0.054	.78	−0.110	.59
HIEE	−0.194	.31	−0.183	.40
TMT-B				
SR-CHx	−0.023	.90	0.079	.70
HIEE	0.269	.16	0.220	.31

Note. β = standardized linear regression coefficient; HVLT-R = Hopkins Verbal Learning Test–Revised; FAS = F-A-S verbal fluency; SDMT = Symbol Digit Modalities Test; TMT-B = Trail-Making Test–Form B. [a]$p = .054$.

Table 4. Relationships between cognitive test raw change scores and exploratory factors. Spearman's Rho (ρ) correlations were calculated between all cognitive test change scores and Brief Symptom Inventory-18 (BSI-18) domains, Pittsburgh Sleep Quality Index (PSQI) Total Score, Alcohol Use Disorders Identification Test (AUDIT) total score, Drug Abuse Screening Test (DAST) total score, and total years of football participation (TYP)

Cognitive test change score	BSI-18 Depression ρ	BSI-18 Anxiety ρ	BSI-18 Somatic ρ	PSQI ρ	AUDIT ρ	DAST ρ	TYP ρ
HVLT-R	0.248	0.032	−0.395	−0.267	−0.085	−0.370	0.021
Immediate Recall	$p = .178$	$p = .863$	$p = .028$[a]	$p = .147$	$p = .649$	$p = .040$[a]	$p = .911$
HVLT-R Delayed	0.273	−0.170	−0.059	−0.119	0.020	−0.056	0.119
Recall	$p = .137$	$p = .360$	$p = .754$	$p = .522$	$p = .915$	$p = .763$	$p = .523$
FAS	−0.235	−0.121	0.059	−0.228	−0.079	−0.145	−0.334
	$p = .203$	$p = .518$	$p = .751$	$p = .217$	$p = .672$	$p = .435$	$p = .067$
SDMT	0.114	−0.144	−0.380	−0.073	−0.055	−0.164	−0.034
	$p = .543$	$p = .438$	$p = .035$[a]	$p = .697$	$p = .768$	$p = .377$	$p = .858$
TMTB	0.199	0.083	−0.005	0.062	0.197	0.284	0.129
	$p = .284$	$p = .655$	$p = .979$	$p = .740$	$p = .288$	$p = .121$	$p = .489$

Note. HVLT-R = Hopkins Verbal Learning Test–Revised; FAS = F-A-S verbal fluency; SDMT = Symbol Digit Modalities Test; TMT-B = Trail-Making Test–Form B. [a]$p < .05$.

measurable effects of American football participation, exposure to head trauma, or other health-related factors, it appears that larger sample sizes and continued longitudinal follow-up will be required to recognize clinically relevant relationships with objectively measured cognitive test performance. Prospective study designs with larger sample sizes that include former players in this early midlife will help account for potential factors that may interact with head trauma exposure or in isolation to affect trajectories of cognitive functioning as former players age through middle and later life phases.

Repetitive head impact exposure

Recent research has sought to examine the relationships between measures of RHIE and performance on objective clinical measures of cognitive functioning among active high school (Brett et al., 2019a), active collegiate (Amadon et al., 2023; Brett et al., 2019a; CARE Consortium Investigators et al., 2019), former high school (Deshpande et al., 2017; Montenigro et al., 2017), former collegiate (Alosco et al., 2023; Brett et al., 2022a; Montenigro et al., 2017; Schaffert et al., 2024), and former professional (Fields et al., 2020; Schaffert et al., 2022; Stamm et al., 2015) American football players. Across these studies, there is mixed evidence for an overall effect of RHIE on chronic, objectively measured cognitive functioning. This lack of consistency may be the result of heterogeneous participant ages, multiple operationalizations of RHIE (e.g., age of first exposure, total years of play, HIEE), and differing clinical measures of cognitive functioning across studies. In the present study, there were no statistically significant associations observed between HIEE and changes in neurocognitive test performances over the study follow-up period. This finding is in line with another recent study in older (50–87 years of age) former collegiate football players using a modified HIEE metric (Schaffert et al., 2024). The HIEE metric was designed to be a more granular approach to estimating RHIE than total participation years alone, knowing that level of play, playing position(s), and team role (e.g., first vs. third

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string) influence time of participation within individual training sessions, games, and seasons (Kerr et al., 2015). However, this tool is limited as it does not account for sport participation prior to high school, which may be captured with other RHIE metrics (e.g., total years of play). Still, the lack of significant association with cognitive test performance changes over time in this sample suggests that there are factors beyond RHIE that need to be accounted for when examining trajectories of cognitive function among former American football players.

Lifetime concussion history

In light of recent work (Kerr et al., 2022b), SR-CHx was operationalized in more than one way in an attempt to avoid failing to detect a potentially significant relationship that might have been influenced by the way this exposure variable was characterized. Neither categorical nor continuously operationalized SR-CHx was observed to have a significant association with any of the cognitive test change scores in this study. The association between continuous SR-CHx and HVLT-R Immediate Recall ($p = .054$) approximated our *a priori* significance threshold, with a medium effect size. Ultimately, we are not able to determine in which direction the *p*-value might shift, if at all, with a larger sample of former collegiate football players. It is possible that the relationship between SR-CHx and changes in HVLT-R Immediate Recall performance may be null. Alternatively, this potential association may indicate a dose–response relationship between more lifetime concussions and worse verbal learning and memory performance while accounting for pre-morbid intelligence and RHIE. Previous studies have indicated that it may not be a pure dose–response relationship and that other factors, such as age and concurrent mood-related symptoms, may interact with concussion history to influence cognition later in life (Gallo et al., 2022; Schaffert et al., 2024; Walton et al., 2022a). As stated previously, there remains a need to better contextualize the many factors that influence trajectories in cognitive functioning in this population beyond concussion history and RHIE alone.

Exploratory factors

In an effort to determine other factors that might affect changes in cognitive test performance in this sample, we explored self-reported symptoms (depression, anxiety, and somatic) and health-related behaviors (sleep quality, alcohol use, drug abuse) at the follow-up time point, as well as another surrogate measure of RHIE during American football (TYP). Greater somatic symptom burden at T2 was related to worse verbal learning and memory (HVLT-R Immediate Recall) and processing speed (SDMT). Within the current sample, it is not clear if endorsement of BSI-18 Somatic scale items reflects physical manifestations of psychological distress (i.e., somatization) or symptoms of physical difficulties and pain. The relationship between bodily symptoms/pain and cognitive dysfunction in former American football players is not well understood; however, concomitant clinical profiles of pain, poor physical functioning, and subjective cognitive difficulties have been observed among former National Football League (NFL) players (Brett et al., 2021b; Roberts et al., 2020). Given the prevalence of ongoing orthopedic issues in this population (Golightly et al., 2009; Grashow et al., 2022; Pietrosimone et al., 2015), links between concussion history and the prevalence of osteoarthritis in former NFL players (Lynall et al., 2017), and the potential for somatic symptomology to moderate of the relationship between concussion history and other clinical outcomes (e.g., depressive symptoms) (Brett et al., 2019b), future research on the long-term effects of head trauma on cognitive function should consider somatic symptoms as another important factor influencing cognitive outcomes. Adding to this complex paradigm, higher DAST scores were related to worse HVLT-R Immediate Recall change scores in our study, and drug abuse has been associated with pain experiences among former football players in a previous study (Mannes et al., 2022). Importantly, we were not able to account for these exploratory factors at T1, nor could we look at the potential for concurrent changes in these factors with changes (or lack thereof) in cognitive test performance over the same time period. It will be important to evaluate trajectories of symptoms, health-related behaviors, and other potentially influential factors in parallel with cognitive functioning in future studies.

Further considerations

Within our sample, over 40% of the participants exhibited low scores on the HVLT-R at T1. While the source of this high rate is unknown, a number of factors may be contributing such as high expectations for performance in healthy, young individuals (small raw score errors result in classification of low score) based on the HVLT-R normative sample, suboptimal engagement during testing, or higher error in proportion of classification status due to smaller sample sizes. At T2, there were far fewer participants exhibiting low scores, which may have been due to a number of factors such as higher education attainment (e.g., cognitive reserve), lifestyle factors (e.g., lifetime participation in exercise), or a number of other unmeasured variables. These disparate performance findings in relation to normative values between T1 and T2, coupled with high inter-individual variability in performance between time points overall, import the need to account for various potentially protective factors (e.g., education attainment, exercise participation) and harmful factors (e.g., drug use, pain) that separate individual trajectories of objectively measurable cognition in former football players. This further supports the need to investigate cognitive test performance in larger samples of former athletes and with multiple time points.

Current and former American football players are understandably concerned about how their cumulative exposure to head trauma from sport may increase risks for poor long-term cognitive outcomes (Alosco et al., 2023; Baugh et al., 2021; Walton et al., 2021b), as well as which factors may represent increased vulnerability for the development of mild cognitive impairment and dementia-related diagnoses (Grashow et al., 2022; Guskiewicz et al., 2005; Walton et al., 2022a), In our related work with an overlapping study sample, we observed a relationship between greater HIEE and smaller hippocampal volumes (Brett et al., 2022b). However, we did not observe a relationship between HIEE and measures of resting-state functional connectivity among the whole brain and large-scale functional networks (Walton et al., 2022b). Given that lower hippocampal volume has been observed as predictive of future incident dementia and cognitive decline in multiple studies

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with non-sport samples (den Heijer et al., 2010; Kantarci et al., 2016), it is possible that decreases in hippocampal volumes may precede later larger-scale connectivity disruption—or both may be required for clinically meaningful functional decline. Further investigation is required with large, longitudinal samples to best understand the relationships among exploratory factors, brain structure, functional connectivity, and trajectories of objectively measurable cognitive functioning in this population. Inclusion of former players at early midlife in ongoing and future research may be vital to understanding the timing of onset for brain health changes that may later be measurable as functional and or neuropathological declines, whether or not measurable changes are related to histories of head trauma.

Limitations

As with all studies that rely on participants to self-report information (e.g., concussion history and HIEE), there may have been errors in the participants' reports. Specific to concussion history, we did not incorporate a structured interview or chart review to corroborate their SR-CHx. Similarly, although the HIEE is an advanced exposure estimation based on available research using helmet sensors at various levels of football play (Kerr et al., 2015), we cannot confirm the veracity of the estimated lifetime RHIE with this tool. Further, HIEE does not account for organized participation before high school, and there may be important RHIE prior to that time frame that was not measured. Of note, we incorporated TYP as a separate, exploratory exposure variable that does account for pre-high school play, and there were no significant associations between TYP and cognitive test change scores in this study. Although participants voluntarily enrolled to be involved in the study, performance validity measures were not employed and non-credible performance on cognitive testing cannot be ruled out. Given the small sample size as well as relative lack of racial/ethnic diversity and other possible determinants of brain health (e.g., education attainment, vocation, socioeconomic status, and lifestyle factors), generalizability of our results to the general population of former collegiate American football players may be limited. Of additional importance is the lack of prospective study of non-football (non-athlete and/or other sport control) participants with which to compare our present findings. Further work with large sample sizes and the ability to account for myriad personal, social, and environmental factors is required to understand the trajectories of brain health in this population.

CONCLUSIONS

In this cohort of former collegiate football players now in early midlife, no significant changes in cognitive test performance were observed over nearly two decades, though the individual variability of these changes was large. While concussion history and RHIE were not related to changes in neurocognitive test performance over 18 years in this sample, multiple exploratory factors were crudely associated with individual test score changes. Our study with a small sample of former football players was an important preliminary step forward in this line of inquiry, and follow-up research with this cohort and larger longitudinal studies are essential to understand the ongoing relationships between head trauma history and cognitive function as former football players age. It is likewise essential to account for the many potential factors that may influence neurocognitive test performance changes over time.

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CONFLICT OF INTEREST

The authorship team reports the following potential conflicts of interest: SRW receives research funding from the United States Department of Defense (DoD), Department of Veterans Affairs (VA), and the National Center for Advancing Translational Science for research projects related to brain injury, previously worked on projects funded by the National Football League (NFL), and currently provides unpaid service to the World Federation of Athletic Training and Therapy (WFATT) and Concussion in Sport Group (CISG). ZYK has previously received funding from The Centers for Disease Control and Prevention (CDC); NFL; National Institutes of Health (NIH); National Operating Committee on Standards for Athletic Equipment (NOCSAE); and United States DoD. JRP is a contractor with General Dynamics Information Technology in support of The National Intrepid Center of Excellence and works on research projects funded by the DoD. KSG has no conflicts of interest to report. MAM reports researching funding to Medical College of Wisconsin from the NIH, Department of Defense, VA, CDC, NFL, National Collegiate Athletic Association (NCAA), and Abbott Laboratories. He previously served as a consultant to Neurotrauma Sciences, Inc., and is a clinical consultant to the Green Bay Packers professional football club. He also reports honoraria and travel support for professional speaking engagements. KMG reports grants from Boston Children's Hospital (sub-award from the NFL). BLB reports grants from the National Institute on Aging and National Institute of Neurological Disorders and Stroke and honoraria for conference presentations.

The authors declare that none of these potential conflicts of interest influenced the present study.

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